

Ethics of AI #1

ECE Paris – Ing5 Data&IA



Farès Fadili

September 25, 2025



Overview of the course

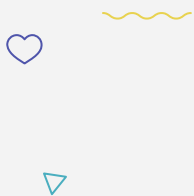


5 two-hours classes, including two practical works and one bonus-graded project :

1. **Culture of AI Ethics** : lecture and group presentations 🙌
2. **Biases** : lecture and practical work
3. **Fine-tuning** : lecture and practical work
4. **XAI & Ethics by design** : lecture ; **Project** : start of group work
5. **Project** : group presentations

Only one “real” grade : **written exam (MCQ)** probably on December 12nd

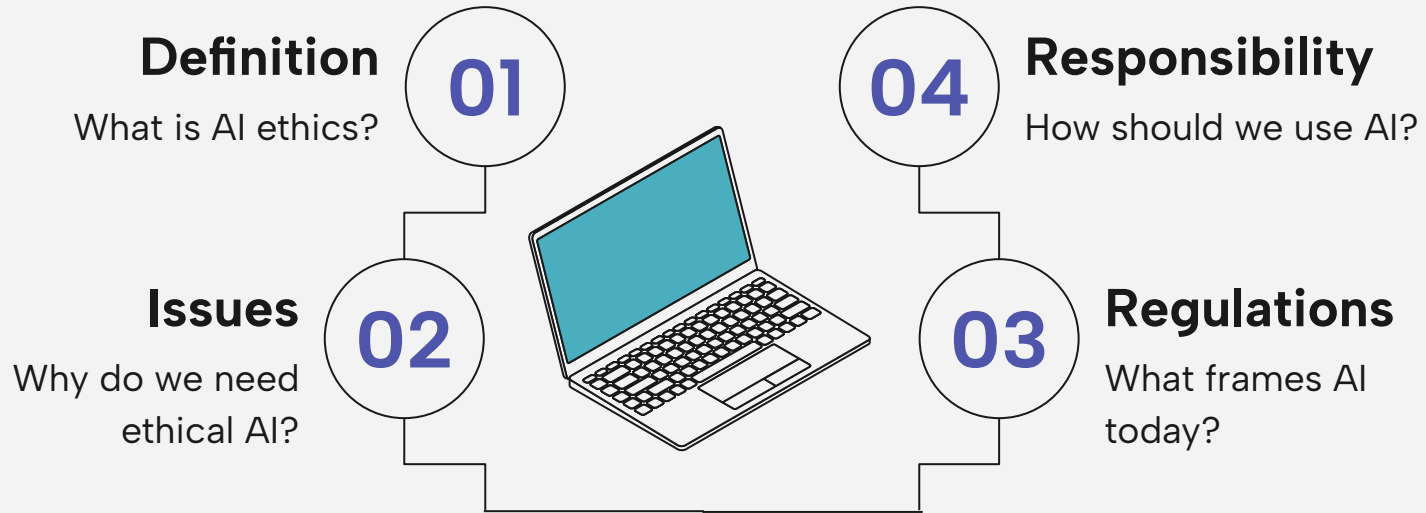




Introduction to AI Ethics

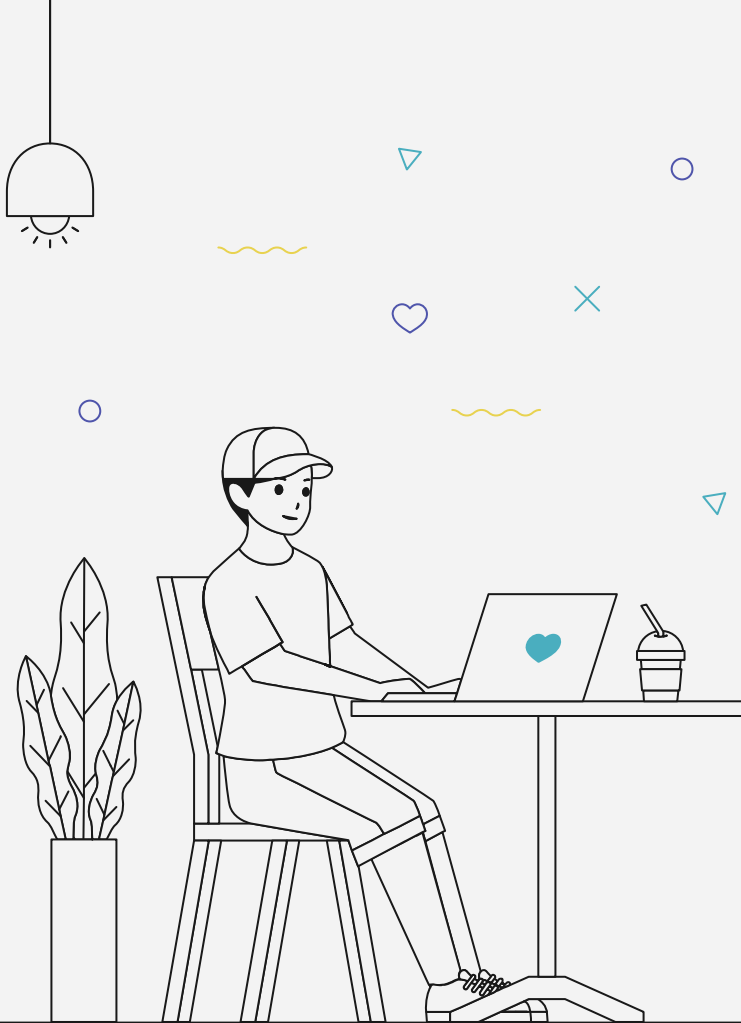


Lecture : contents



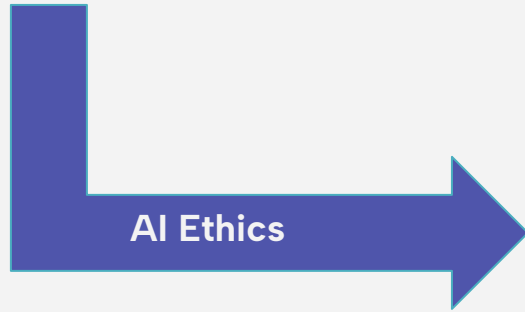
01 Definition

What is AI Ethics?



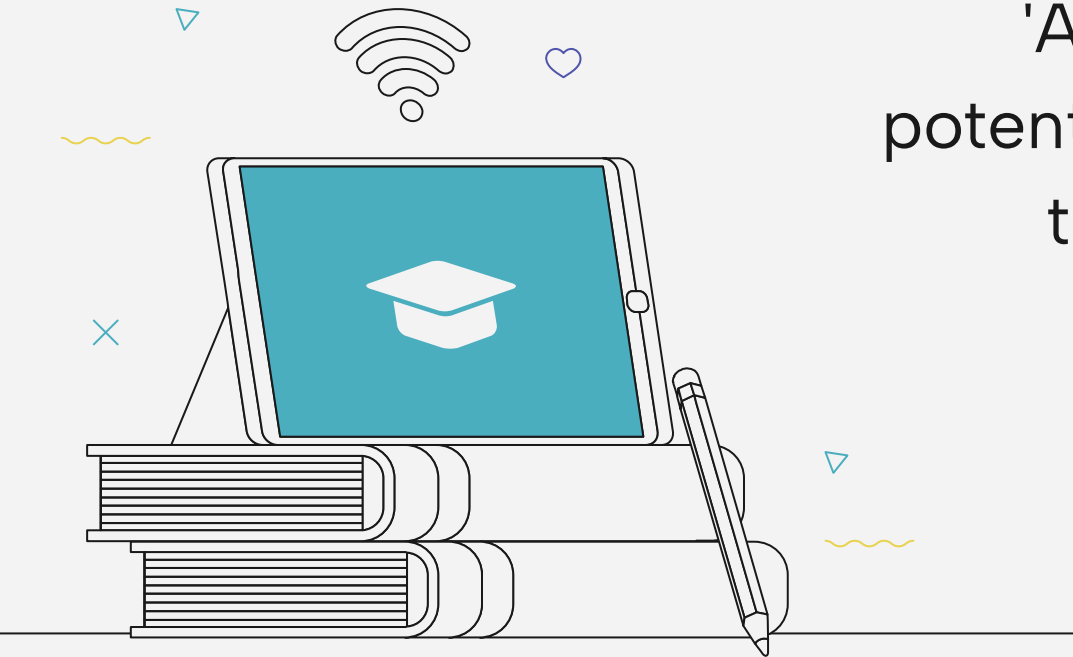
What is AI Ethics?

Ethics: The set of moral principles that underpin a person's actions.



Principles and values guiding the development and use of AI systems to ensure they respect human rights, fairness, and the common good.





'Artificial intelligence is
potentially more dangerous
than nuclear weapons'

— **Sam Altman**

02 Issues

Why do we need ethical AI? ♡





Issue 1: Privacy

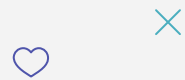
Issue: AI systems collect and store sensitive data, raising questions about privacy.

Example: Data breaches, such as Cambridge Analytica, have exposed the personal data of millions of Facebook users.

Solution: Strengthen data protection regulations, such as the RGPD in Europe, to ensure that :

- users have control over their data.
- companies and institutions adhere to strict security and privacy standards.





Issue 2: Discrimination



Problem: Biases in training data and AI algorithms can lead to unfair discrimination.

Example: Automated credit rating systems may favor certain categories of the population over others due to biases in the data.

Solutions:

- Conduct regular audits of data used in AI systems to identify and correct biases.
- Promote diversity when collecting data and building AI development teams to reduce bias.





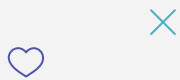
Issue 3: Transparency

Problem: The opacity of decisions made by AI systems makes it difficult to understand understanding how they work.

Example: Deep learning models such as deep neural networks are often difficult to explain in simple terms.

Solution: Develop techniques and tools (XAI) to explain the results of AI models, such as the granting or refusal of a bank loan.





Issue 4: Security



Problem: Vulnerabilities in AI systems can be exploited for malicious purposes.

Example: Attackers can use adversarial attacks to trick computer vision systems into making erroneous decisions.

Solutions:

- Strengthen the security of AI systems by adopting secure development practices and conducting regular security audits.
- Raise awareness of AI security and promote sound cybersecurity practices to prevent abuse and malicious attacks.





Issue 5: Employment



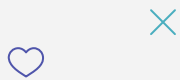
Issue: Automation through AI can lead to job losses, raising questions about social protection and worker training.

Example: Cashiers in supermarkets are being replaced by automatic checkouts, which may lead to job losses.

Solutions:

- Invest in training and skills development to help workers adapt to the changes caused by AI automation.
- Implement social protection policies to support workers affected by automation.





Issue 6: Environment and society



Issue: The growing use of AI has an impact on society and the environment, raising questions of social and environmental responsibility.

Example: The carbon footprint of the data centers required to power AI systems is significant.

Solution:

- Encourage the adoption of sustainable development practices in AI, such as optimizing infrastructure energy and reducing carbon footprints.
- Promote corporate social responsibility by integrating ethical, social and environmental considerations into AI-related decisions.





Issue 7: Autonomy and control

Problem: Autonomous AI systems can make decisions without human intervention, which can lead to damage or harm if decisions are incorrect or unfair.

Example: An autonomous AI system used to control a surgical robot may cause injury to a patient if the system is not functioning properly.

Solution: Human-supervision mechanisms, which enable humans to monitor the decisions made by AI systems, and intervene if necessary.



12

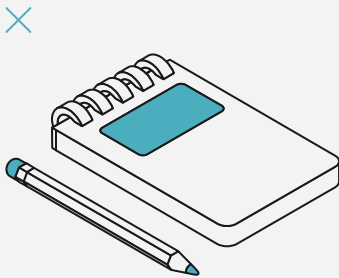
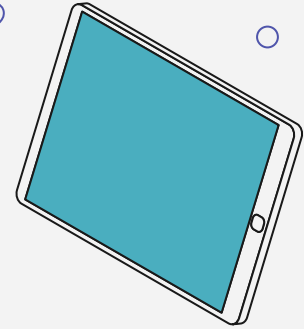
Number of seconds to create a deepfake

70,000,000

Annual CO2 emissions for driving models

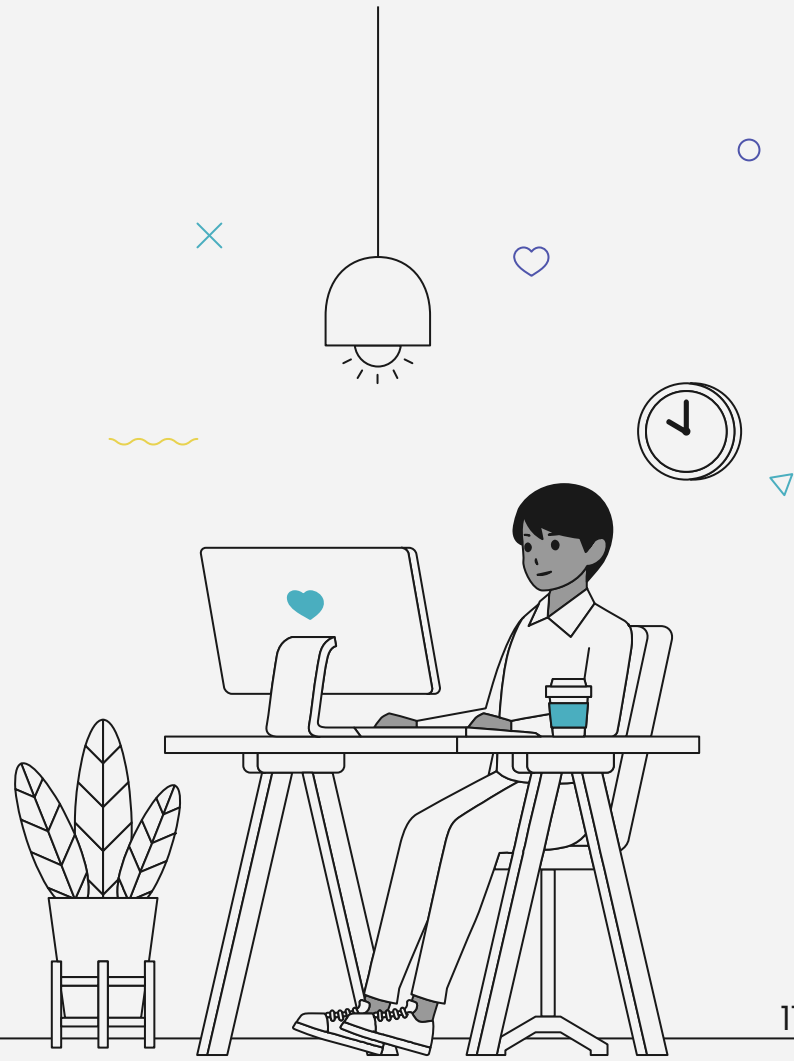
2,500,000,000

Number of personal data exposed via AI



03 Regulations

What frames AI today?





Major initiatives



RGPD (European Union, 2018): Regulates the collection and use of personal data, essential for AI systems.

AI Act (European project, 2025): Classification of AIs according to their risks (unacceptable, high, limited, minimal) with specific requirements.

Algorithm Act of 2021 (United States): Requires transparency in algorithmic decisions with significant public impact.

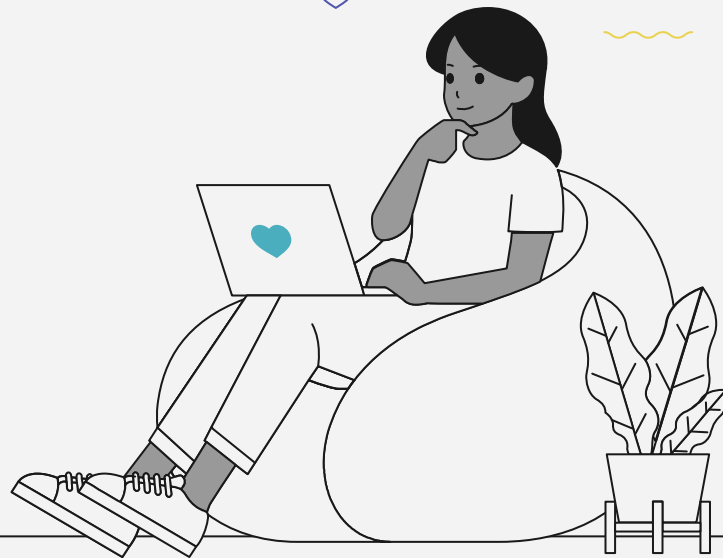
UNESCO Initiative (2021): Global ethical recommendations on AI transparency, fairness and sustainability.



04

Responsibilities

How should we use AI?





Responsible use of AI

01

Step back

Understanding the limits of the system

03

Critique

Questioning results

05

Awareness

Promoting responsible use around us

02

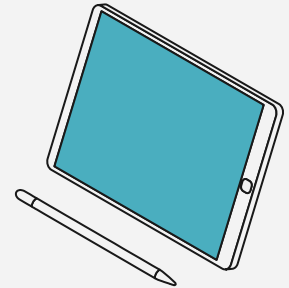
Conscience

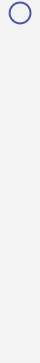
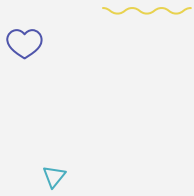
Employing ethically, respecting rights and privacy

04

Reporting

Report biases and negative impacts





Group presentations





Group presentations

Teams: 6 students

Duration: 10 minutes per group

Objective: Present an AI application and analyze its ethical implications using concepts from today's lecture.

Suggested Structure:

AI Solution (2 min): Brief description

Ethical Issues (6 min): Analyze 3–4 key issues (privacy, security, transparency, etc.)

Regulations (2 min): Identify relevant frameworks (GDPR, AI Act, etc.)

Conclusion (1 min): Key takeaways

